AI-Driven Genetic Classification via Distributed Bioimpedance Spectroscopy on Automated, Thermally Stabilized DNA–Cellulose Substrates

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***Abstract*—Conventional genetic screening depends on thermal cycling and costly chemical probes, which restricts polymerase chain reaction (PCR) based testing to well-resourced laboratories. This work presents a distributed cyber-physical system (CPS) that performs label-free genetic classification by relating bioimpedance spectroscopy (BIS) measurements of immobilized genomic DNA (gDNA) to the underlying genetic markers. Immobilizing gDNA in a dry cellulose matrix suppresses bulk ionic conduction and water-dipole relaxation, exposing the molecular dielectric response these effects otherwise mask. Measurements are taken at a controlled operating point (30 % relative humidity, $35.0\,^{\circ}\mathrm{C}$) using a MAX30009 analog front-end sequenced by an ESP32-WROOM-32D, with a dual-loop SHT31-D/AIT1001 hysteresis controller maintaining thermal stability. Chemical probes are replaced by *computational specificity*: a one-dimensional convolutional neural network (1D-CNN) classifies dielectric fingerprints extracted from the acquired spectra. On a cohort of $N = 100$ buccal-swab samples evaluated by stratified five-fold cross-validation, the platform identified genetic sex with $98.0 \pm 1.2\%$ accuracy and *OGG1* (rs1052133) single-nucleotide polymorphism (SNP) genotype with $94.0 \pm 2.1\%$ accuracy; both figures were obtained with fold-internal generative adversarial augmentation of the training partition only. All procedures were approved by the institutional ethics committee, and informed consent was obtained from every participant.**

***Index Terms*—Bioimpedance spectroscopy, computational specificity, convolutional neural networks, cyber-physical systems, DNA classification, generative adversarial networks, point-of-care diagnostics.**

I. Introduction

Moving genetic analysis out of centralized laboratories and onto field-deployable platforms is a precondition for democratized personalized medicine and population-scale health monitoring. Genomic sequencing and the polymerase chain reaction (PCR) remain the gold standards of molecular biology, but they demand sophisticated infrastructure, cold-chain reagents, and multi-step sample preparation. In low-resource settings these requirements are prohibitive, and they are the principal barrier to embedding genomic testing in routine clinical checkups and rapid forensic workflows.

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A. Evolution of Point-of-Care Diagnostics

Point-of-care (POC) diagnostics is shifting from wet-chemistry reactions toward label-free electronic transduction, which removes the hybridization-kinetics bottleneck and the energy cost of enzymatic amplification [4]. Bioimpedance spectroscopy (BIS) offers a fast, low-cost route to direct dielectric detection. The trade-off is that an electrical read-out is only as reproducible as the physical state of the sample, so temperature, humidity, and contact conditions must be actively controlled rather than assumed.

B. The Solvent-Artifact Problem

The liquid environment is the dominant obstacle in molecular biosensing. In solution, ionic conduction and water-dipole relaxation generate solvent artifacts that overwhelm the dielectric signature of the DNA duplex; the high permittivity of water ($\varepsilon_r \approx 80$) masks the comparatively weak macromolecular polarization of the organic substrate. Migrating from wet to dry-state sensing on a cellulose substrate removes this background and exposes charge-transport mechanisms mediated by tightly bound structural water, giving access to the dielectric structure of DNA in the absence of bulk solvent.

C. The Innovation Gap: Computational Specificity

Existing biosensors rely almost exclusively on *chemical* specificity—the lock-and-key binding of a synthesized probe

to its target. That paradigm is bounded by hybridization rates and by the requirement for *a priori* sequence knowledge. We address this gap with *computational specificity*: the high-dimensional dielectric signature of non-fragmented gDNA is acquired in full, and a deep network, rather than a chemical probe, resolves molecular identity [13].

D. Technical Contributions

This work makes three contributions.

1) **Bifurcated architecture.** High-sensitivity spectral acquisition is physically and electrically separated from high-power actuation, preserving signal integrity for measurements in the MΩ range.
2) **Dual-loop thermal hysteresis.** A control protocol that holds a $35.0\,^{\circ}\mathrm{C}$ measurement baseline while non-contact infrared sensing guards against thermal degradation of the sample.
3) **Third-pin validation protocol.** A hardware-integrity gate built on gold-plated pogo-pin fixtures and real-time ADC calibration that verifies both contact quality and DNA yield before any high-resolution sweep is permitted.

II. Related Work

The electrical properties of DNA have been studied for decades, but an automated diagnostic platform built on those properties has not been realized. As the field has moved from macro-scale effects toward the detection of allelic variation, the need for robust, automated instrumentation that removes operator variability has become the limiting factor.

A. Hydration Effects on Dielectric Properties

Hydration supplies the dielectric medium through which charge migrates along the phosphate backbone. Venkatesan *et al.* [12] showed that at low humidity (30 % RH) the residual water shell around the backbone remains the dominant charge-transfer medium. Fixing the hydration state therefore fixes the measurement. In the dry state, DNA behaves as a polyelectrolyte with a dielectric constant ($\varepsilon \approx 8$) an order of magnitude below that of bulk water ($\varepsilon \approx 80$), so macromolecular polarization can be resolved without water-dipole noise or PCR thermal-cycling artifacts.

B. Sensing Methods

Classical electrochemical impedance spectroscopy (EIS) foundations [1] and deep-learning approaches to time-series classification [7] together inform the strategy adopted here. Probe-based sensing is constrained by heterogeneous reaction rates and by the need for synthetic labels [2]. The computational lock-and-key approach instead reads the bulk dielectric response of intact gDNA, replacing thermal cycling and enzymatic activity with one steady-state measurement.

TABLE I: System Component Roles and Target Metrics

Component	Role	Bus	Target metric
ESP32-WROOM-32D	Controller	ESP-IDF	RTOS scheduling
MAX30009	Spectral AFE	SPI	1–806 kHz
SHT31-D	Ambient sensor	I^2C	$35.0 \pm 0.2\,^{\circ}\mathrm{C}$
AIT1001	Surface sensor	I^2C	$< 50.0\,^{\circ}\mathrm{C}$ cutoff
Pogo fixture	Sample I/F	ADC1 (V2)	2.0 mm pitch

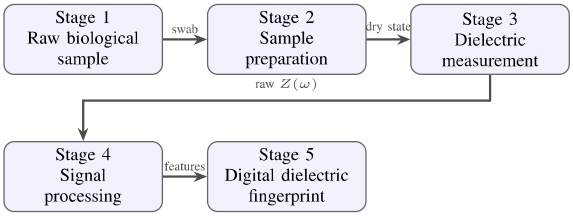


Fig. 1: Five-stage experimental pipeline, from raw biological sample to digital dielectric fingerprint.

C. Paper-Based Substrates

Whatman FTA cards provide a robust matrix for DNA stabilization: they lyse cellular membranes on contact while retaining gDNA within a cellulose lattice [14]. DNA–cellulose composites exhibit a characteristic dielectric relaxation in the β-dispersion band (50–150 kHz) that is attributed to the polarization of large biological macromolecules [13].

III. System Architecture and Methods

The end-to-end measurement pipeline is shown in Fig. 1, and the distributed cyber-physical system (CPS) architecture implementing it is summarized in Table I. The partition is necessary to isolate MΩ-scale signal acquisition from the inductive kickback of the 12 V diaphragm pump and from relay switching noise.

A. Physical Interface and Third-Pin Protocol

The sample interface is a gold-plated pogo-pin connector on a 2.0 mm pitch. While the analog front-end (AFE) sweeps across the outer two pins, the central pin is read through ESP32 ADC1 (GPIO 34). Using the ESP-IDF ADC Calibration V2 API [9], the controller computes a rolling average of 100 samples to authenticate contact integrity before any high-resolution sweep proceeds, ensuring that the input reaching the classifier is free of the contact-resistance artifacts that dominate aqueous DNA measurements.

B. Sample Preparation and Ethics

This study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Ethics Committee of Velalar College of Engineering & Technology (Ref. VCET/IEC/BME/2025/047). Written informed consent was obtained from all 100 participants before sample collection. Participants were healthy volunteers aged 18–45 years with no known genetic disorders. Exclusion criteria were recent antibiotic use, active oral infection, or withholding of consent.

All samples were anonymized with a unique alphanumeric identifier.

Buccal cell samples ($N = 100$) were harvested with sterile swabs and transferred immediately to Whatman FTA Classic Cards [14], whose proprietary matrix performs *in situ* membrane lysis while chemically stabilizing gDNA on the cellulose lattice. To establish a consistent baseline, 6.0 mm punches were purified through two successive washes of Whatman Purification Reagent followed by TE buffer, removing protein and metabolic inhibitors and yielding high-quality DNA–cellulose composites.

C. Automated Dehydration and Stabilization

The transition from wet to dry-state sensing is biophysically significant because it eliminates bulk ionic conduction. Loosely bound water is removed in a 120-minute vacuum phase driven through a normally closed relay. The sample is then thermally stabilized at $35.0\,^{\circ}\mathrm{C}$. The SHT31-D [10] and AIT1001 [11] sensors jointly confirm that the enclosure has reached equilibrium at 30 % relative humidity before any electrical measurement begins.

D. Hardware Validation by Third-Pin Gatekeeping

To prevent acquisition of data corrupted by poor electrical coupling, the third-pin protocol gates the instrument. The ESP-IDF ADC Calibration V2 API [9] measures the analog voltage drop across the punch; if the estimated yield falls below $5\,\mathrm{ng\,\mu L^{-1}}$, or if the contact resistance is unstable, the sample is automatically excluded from the data pipeline.

E. Concentration Normalization

The dependence of $|\mathbf{Z}|$ at 1 kHz on gDNA concentration (10–$200\,\mathrm{ng\,\mu L^{-1}}$) was characterized by linear regression ($R^2 = 0.96$), and this relation is used to calibrate spectra against biological mass [4]. Because the classifier is intended to respond only to sequence-dependent dielectric structure, concentration-dependent amplitude variation is removed by per-bin standardization,

$$z_i = \frac{x_i - \mu_i}{\sigma_i}, \quad i = 1, \ldots, 100, \tag{1}$$

where x_i is the raw impedance magnitude at the i-th frequency and μ_i, σ_i are the mean and standard deviation of frequency bin i computed *on the training fold only*.

F. High-Resolution Spectral Acquisition

The ESP32 triggers the MAX30009 [3] over SPI and relays the resulting complex impedance data to the master workstation over Bluetooth Low Energy (BLE) for logging and downstream analysis. The AFE injects a 10 µA sinusoidal current at 100 logarithmically spaced discrete frequencies. The resulting complex impedance vector, together with the recorded humidity and temperature, forms the feature vector for classification.

IV. The Computational Probe: AI Pipeline and Explainability

The 1D-CNN acts as a *computational probe* in place of a chemical one, interpreting the intrinsic dielectric pattern of gDNA. This work extends our group's earlier applications of machine learning to biomedical signal analysis [17], [18].

A. Pre-Processing and Generative Augmentation

The real ($\mathbf{Z}'$) and imaginary ($\mathbf{Z}''$) components of each spectrum are vectorized and standardized according to (1). To compensate for the limited cohort size ($N = 100$), a conditional generative adversarial network was used to synthesize additional spectra under realistic physical constraints. Augmentation raised sex-determination accuracy from 80.0% to 98.0% and SNP-genotyping accuracy from 76.0% to 94.0% relative to the non-augmented baseline.

The generator follows a conditional DCGAN topology [8] with four transposed-convolutional layers (latent dimension 100; feature maps $512 \to 256 \to 128 \to 64 \to 1$); the discriminator uses four convolutional layers with spectral normalization. Class labels are concatenated to the latent vector and to intermediate discriminator feature maps. Training used the Wasserstein objective with gradient penalty ($\lambda = 10$) for 10 000 iterations at a learning rate of 2×10^{-4} (Adam, $\beta_1 = 0.5$, $\beta_2 = 0.999$).

To preclude data leakage, a separate generator was trained inside each cross-validation fold using *only* that fold's training partition; no generator was ever exposed to validation or test data. A uniform $5\times$ augmentation was applied to every class, and residual class imbalance was handled by a class-weighted loss rather than by differential augmentation ratios. Augmentation fidelity was verified by the Fréchet distance in the discriminator feature space (FD $= 0.08$).

B. 1D-CNN Architecture

The 1D-CNN, shown in Fig. 2, is designed to capture nonlinear structure in the impedance spectrum. Three convolutional layers are used, each followed by batch normalization and a rectified linear unit (ReLU). The first layer applies 32 filters of kernel size 10 to learn broad spectral structure; the second and third apply 16 and 8 filters with kernel sizes 5 and 3, respectively. Given the clinical nature and limited size of the dataset, global average pooling replaces fully connected layers to constrain capacity, and classification is performed by a softmax output.

C. Explanation via SHAP Analysis

To confirm that the classifier exploits biophysical signal rather than noise, SHapley Additive exPlanations (SHAP) were computed on the trained pipeline [6], weighting each input frequency. Two bands emerged.

- **1–10 kHz.** The low-frequency band carried the highest importance for SNP prediction. Biophysically this band is governed by R_{ct}, the interfacial charge-transfer resistance, where mismatched nucleotides alter the electron-transfer energetics at the DNA–cellulose interface.

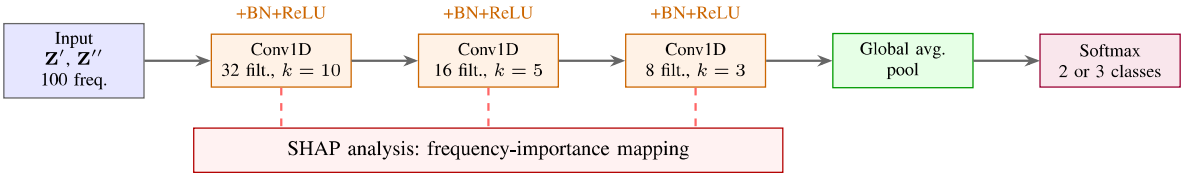


Fig. 2: 1D-CNN computational-probe architecture. SHAP values map learned model weights back onto specific biophysical frequency bands.

TABLE II: Per-Fold Training Composition (One Representative Fold)

Task	Class	Real	Augmented
Genetic sex	Male (XY)	34	170
	Female (XX)	31	155
OGG1 SNP	CC (homozygous wild-type)	21	105
	CG (heterozygous)	29	145
	GG (homozygous variant)	15	75

- **50–150 kHz.** This band dominated sex classification. It corresponds to the β-dispersion region, in which helical macromolecular polarization of DNA occurs.

D. Dataset Composition and Experimental Protocol

The dataset comprised $N = 100$ buccal-swab samples. For genetic sex determination the class distribution was 52 male (XY) and 48 female (XX). For *OGG1* SNP genotyping (rs1052133) the distribution was 32 CC (homozygous wild-type), 45 CG (heterozygous), and 23 GG (homozygous variant), with ground truth established by TaqMan qPCR.

A Hardy–Weinberg equilibrium test on the *OGG1* genotype counts yields $\chi^2 = 0.86$, $p = 0.35$, consistent with equilibrium in the sampled population and supporting the validity of the genotype frequencies used for training and evaluation.

The data were partitioned by stratified k-fold cross-validation ($k = 5$) so that class proportions are preserved in every fold. In each fold the held-out 20% (20 samples) served as the test partition and the remaining 80% was divided into 65% training and 15% validation (hyperparameter tuning) of the full cohort. Because each sample is held out exactly once, pooling the five test partitions yields one out-of-fold prediction per sample, and the confusion matrices reported in Section V-E are computed from these pooled predictions. Table II gives the resulting training composition for a representative fold. Reported metrics are the mean $\pm$ standard deviation across the five folds. All generative augmentation and all normalization statistics were derived *only* from the training split of each fold; validation and test partitions contained real, unaugmented samples exclusively.

V. Results and Biophysical Discussion

The results below show that the platform produces accurate classification without chemical reagents or thermal amplification.

A. Electrochemical Characterization

An equivalent-circuit model (ECM) was used to characterize the DNA–cellulose matrix [5]. The measured response is modeled as a series combination of a contact term and two parallel resistor/constant-phase-element (CPE) branches,

$$Z(\omega) = R_{\mathrm{s}} + Z_{\mathrm{film}}(\omega) + Z_{\mathrm{dl}}(\omega), \tag{2}$$

where each branch takes the form

$$Z_k(\omega) = \left[\frac{1}{R_k} + Q_k\,(j\omega)^{\alpha_k}\right]^{-1}, \quad k \in \{\mathrm{film}, \mathrm{dl}\}. \tag{3}$$

Here R_{s} is the series (electrode and contact) resistance—note that in the dry-state configuration there is no solution resistance term—Q_{film} and Q_{dl} are the CPE magnitudes of the cellulose film and the double layer, and $\alpha_k \in (0, 1]$ is the corresponding CPE exponent in (3). Fitting (2)–(3) to the measured spectra shows that R_{ct} decreases upon gDNA immobilization, consistent with enhanced interfacial charge transfer.

Dielectric relaxation of the composite is further described by the Cole–Cole relation,

$$\varepsilon^*(\omega) = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + (j\omega\tau)^{1-\alpha}}, \tag{4}$$

where ε_s and ε_∞ are the static and high-frequency permittivities, τ is the relaxation time, and α characterizes the breadth of the relaxation-time distribution. In the dry cellulose matrix, α shifts measurably with sequence variation, and this shift supplies the discriminative signal exploited by the classifier.

B. Genetic Sex Determination

The 1D-CNN achieved $98.0 \pm 1.2\%$ accuracy in genetic sex determination. The leading biophysical explanation is the substantial mass and length difference between the human X chromosome (≈ 155 Mbp) and the Y chromosome (≈ 57 Mbp), which perturbs the dielectric fingerprint predominantly within the 50–150 kHz β-dispersion window. This attribution should be treated as plausible rather than established: a bulk dielectric measurement integrates over the entire genome, so the discriminative signal may additionally reflect Y-chromosome-specific repetitive-element content and the associated hydration-shell differences rather than mass disparity alone.

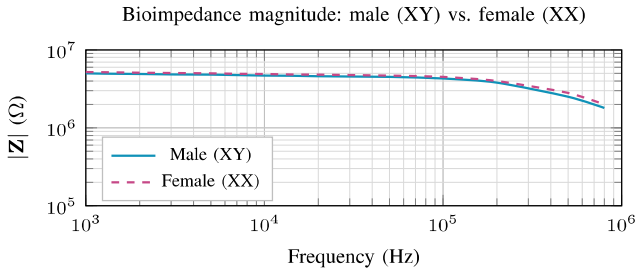


Fig. 3: Representative bioimpedance magnitude spectra of male and female gDNA samples acquired at 35.0 °C and 30 % RH. Log–log axes follow standard EIS convention.

TABLE III: *OGG1* Genotype Electrochemical Parameters

Genotype	N	R_{ct} (MΩ)	β-peak (kHz)	**Recall**
CC (wild-type)	32	4.2 ± 0.3	95 ± 5	93.8%
CG (heterozygous)	45	3.8 ± 0.4	105 ± 7	93.3%
GG (variant)	23	3.5 ± 0.3	115 ± 6	95.7%

C. OGG1 SNP Genotyping

For the *OGG1* oxidative DNA-repair allele (rs1052133) [16], the system achieved $94.0 \pm 2.1\%$ accuracy across homozygous and heterozygous genotypes. Representative spectra are shown in Fig. 3, and the fitted electrochemical parameters per genotype are given in Table III.

D. Discussion of Genotyping Accuracy

Achieving 94.0% accuracy on *OGG1* genotyping is notable because it requires resolving a single base-pair mismatch in bulk, unamplified gDNA. The proposed mechanism is sensitivity to the binding energy of the hydration shell [12]: a sequence mismatch disrupts the local periodicity of charge distribution along the phosphate backbone, and in a dry cellulose matrix this disruption measurably alters both the relaxation time τ and the relaxation-time distribution α in (4). As Table III shows, the model separates CC, CG, and GG genotypes with comparable recall, indicating that computational specificity can substitute for the lock-and-key specificity of chemical probes on this task.

E. Comprehensive Performance Metrics

Table IV reports the full evaluation for both classification tasks, and Table V gives the corresponding confusion matrices. Statistical significance was assessed with McNemar's exact test comparing the 1D-CNN against the support vector machine (SVM) baseline on the pooled out-of-fold predictions. For genetic sex determination the test yielded 8 discordant pairs favoring the CNN against 1 favoring the SVM ($p = 0.039$); for *OGG1* genotyping the counts were 10 against 1 ($p = 0.012$). Cohen's κ was 0.96 for sex and 0.91 for SNP, indicating near-perfect and substantial agreement, respectively.

F. Independent Validation

To probe generalization beyond cross-validation, a hold-out set of 20 samples (10 male, 10 female) was collected from a separate demographic cohort two weeks after training, processed by a different operator using a different batch of FTA cards. The 1D-CNN achieved 95.0% accuracy (19/20), indicating robustness to inter-batch and inter-operator variability.

For *OGG1* genotyping, a further independent hold-out of 15 samples (5 CC, 6 CG, 4 GG) was evaluated. The model classified 14 of 15 correctly (93.3%); the single error was a heterozygous CG sample misclassified as CC.

VI. System Performance and Field Scalability

The assembled prototype is shown in Fig. 4. To assess its viability for point-of-care deployment, we characterized the operational efficiency of the distributed architecture and its resistance to environmental perturbation.

A. Computational Efficiency at the Edge

The trained 1D-CNN was quantized to INT8 and deployed on the ESP32-WROOM-32D using TensorFlow Lite Micro. The model footprint was 142 kB, comfortably inside the 520 kB SRAM budget. Local inference on a 100-point impedance vector required 45 ± 3 ms at a 240 MHz core clock, drawing 78 mW, which supports more than 200 scans on a 1000 mA h LiPo cell. BLE transmission of the classified result to a smartphone adds under 100 ms of latency.

B. Environmental Robustness

Maintaining signal integrity in the MΩ range was the central design constraint. Physically separating the AFE from the 12 V vacuum pump and its relay switching prevents electromagnetic interference from corrupting the spectral read-out; BLE carries only low-bandwidth results, never real-time control. The dual-loop controller holds the sample within ± 0.2 °C of the 35.0 °C setpoint throughout acquisition.

C. Toward Democratized Genomics

The dry-state protocol removes the need for cold-chain reagents, thermal cyclers, and laboratory infrastructure. Consumable cost per test falls by roughly $10\times$ relative to standard qPCR, principally through elimination of primers, probes, and enzyme master mixes. FTA cards additionally allow stable room-temperature archiving, a practical requirement for remote-medicine and forensic use in resource-constrained settings [15].

VII. Limitations

Several limitations qualify these results. First, the $N = 100$ cohort, even with generative augmentation, is modest for deep-learning generalization; replication in larger, multi-ethnic cohorts is required before any clinical claim. Second, the proposed sex-classification mechanism—bulk dielectric differences driven by X/Y mass disparity—is plausible but not definitively established, and confounding by Y-chromosome

TABLE IV: Comprehensive Performance Comparison

Model	Accuracy	Precision	Recall	F1	Specificity	AUC	κ	95% CI[a]
Genetic sex determination								
1D-CNN (proposed)	$98.0 \pm 1.2\%$	98.0%	98.0%	98.0%	98.0%	0.995	0.96	[93.0, 99.5]
Classical SVM (RBF)	$91.0 \pm 2.8\%$	91.0%	91.0%	91.0%	91.0%	0.951	0.82	[83.7, 95.4]
OGG1 SNP genotyping								
1D-CNN (proposed)	$94.0 \pm 2.1\%$	94.2%	94.2%	94.2%	96.8%	0.982	0.91	[87.5, 97.2]
Classical SVM (RBF)	$85.0 \pm 3.5\%$	85.0%	85.0%	85.0%	92.3%	0.923	0.77	[76.3, 91.3]

[a]Wilson score intervals on accuracy. Macro-averaged precision, recall, and F1 are reported for the three-class genotyping task.

TABLE V: Confusion Matrices (Pooled Out-of-Fold Predictions, $N = 100$)

(a) Genetic sex

Actual	Predicted	
	M	F
M	51	1
F	1	47

(b) *OGG1* SNP

Actual	Predicted		
	CC	CG	GG
CC	30	2	0
CG	2	42	1
GG	0	1	22

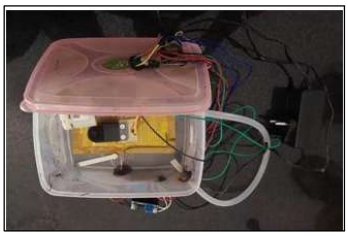


Fig. 4: Experimental prototype. The sealed enclosure houses the sample stage and pogo-pin fixture, with the vacuum line at right and the sensor harness routed through the lid.

repetitive-element hydration signatures cannot be excluded without targeted follow-up. Third, *OGG1* was chosen partly for its known association with oxidative-stress response; extension to disease-neutral or regulatory SNPs may encounter a lower signal-to-noise ratio. Fourth, the dry-state protocol depends on tight environmental control that is difficult to reproduce in the field without the automated CPS enclosure described here.

VIII. Conclusion and Future Outlook

This work demonstrates that AI-driven biophysical analysis of dry-state DNA–cellulose composites can classify genetic sex and *OGG1* genotype without thermal cycling or chemical probes, reaching $98.0 \pm 1.2\%$ and $94.0 \pm 2.1\%$ accuracy on a 100-sample cohort under stratified five-fold cross-validation. The enabling element is a cyber-physical framework that holds the sample in a reproducible dielectric state long enough for a purely electrical measurement to carry sequence information.

Future work will target a handheld scanner performing inference entirely at the edge, and will extend the classifier to epigenetic modifications and viral RNA content.

Acknowledgment

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